IEEE Tables for Meeting Analyzer

Standalone manuscript-ready tables based on recorded implementation and evaluation data

These tables are separated from the evaluation-graphs PDF so they can be inserted independently into the IEEE manuscript. Jaccard-overlap results are intentionally excluded.

TABLE I. SYSTEM TECHNOLOGY STACK

Layer	Technology	Role
Frontend	React, Vite, Tailwind CSS	Interactive dashboard
Backend	FastAPI, Python	API and orchestration
Audio	FFmpeg	Extraction and normalization
Speech recognition	Faster-Whisper	Timestamped transcription
NLP	spaCy, NLTK	Preprocessing and linguistic analysis
Classification	TF-IDF + Logistic Regression	Dialogue-act classification
Topics	LDA / NMF	Topic discovery
Storage	Local JSON / Supabase	Persistence and recovery
PDF export	ReportLab	Report generation

TABLE II. DATASET AND EVALUATION CONFIGURATION

Parameter	Value
Source recordings	1
Clip durations	30 s, 60 s, 120 s
Transcription repeats	3 per duration
Re-analysis runs	5
Whisper model	Systran/faster-whisper-base
Device / compute	CPU / int8
Beam size	5
VAD	Enabled; 500 ms minimum silence
Language	Automatic detection

TABLE III. TRANSCRIPTION LATENCY UNDER CPU/INT8 EXECUTION

Duration (s)	Mean (s)	SD (s)	Minimum (s)	Maximum (s)	Mean RTF
30	2.315	0.483	1.757	2.614	0.077
60	5.715	0.563	5.065	6.041	0.095
120	11.562	0.537	11.063	12.130	0.096

TABLE IV. CORE EXTRACTION COUNTS ACROSS FIVE FRESH ANALYSES

Category	Run 1	Run 2	Run 3	Run 4	Run 5
Action items	2	2	2	2	2
Decisions	1	1	1	1	1
Questions	1	1	1	1	1
Follow-ups	0	0	0	0	0

Follow-up extraction accuracy is not established by this fixture because it contains no separate follow-up item.

TABLE V. VALIDATION PIPELINE RETENTION

Category	Generated	Parsed	Validated	Evidence matched	Deduplicated	Persisted
Action items	2	2	2	2	2	2
Decisions	1	1	1	1	1	1
Questions	1	1	1	1	1	1
Follow-ups	0	0	0	0	0	0

TABLE VI. SUMMARIZATION METHODS

Method	Algorithm / model	Output	Grounding
Extractive	TF-IDF + centroid similarity + MMR	Source excerpts	Direct transcript text
Abstractive	BART/T5-compatible Transformer	Generated draft	Requires human review
Semantic	Structured evidence extraction	Evidence-linked bullets	Verbatim evidence required

TABLE VII. LIMITATIONS AND SAFEGUARDS

Risk	Safeguard	Remaining limitation
Unsupported insight	Verbatim evidence matching	Lexical matching is not full entailment
Re-analysis drift	Versioned manifest and stable IDs	Fixture-level validation only
Speaker identity	Preserve available labels	No diarization for unlabeled audio
Long transcripts	Bounded chunks and deterministic merge	Higher resource requirements
Abstractive hallucination	Extractive default and review	Generated drafts need human review

Data provenance

Tables I and VI reflect the implemented system design. Tables II–V use the recorded benchmark and consistency-validation artifacts in the repository. The latency benchmark uses one source recording, three clip durations, and three repeats per duration. The extraction and retention tables describe a five-run meeting fixture and should not be interpreted as corpus-wide accuracy claims.